CSE_3523

8086 Microprocessor Architecture

Microprocessor, microcontroller and embedded system

Course code: CSE-3523

Credit Hour: 3

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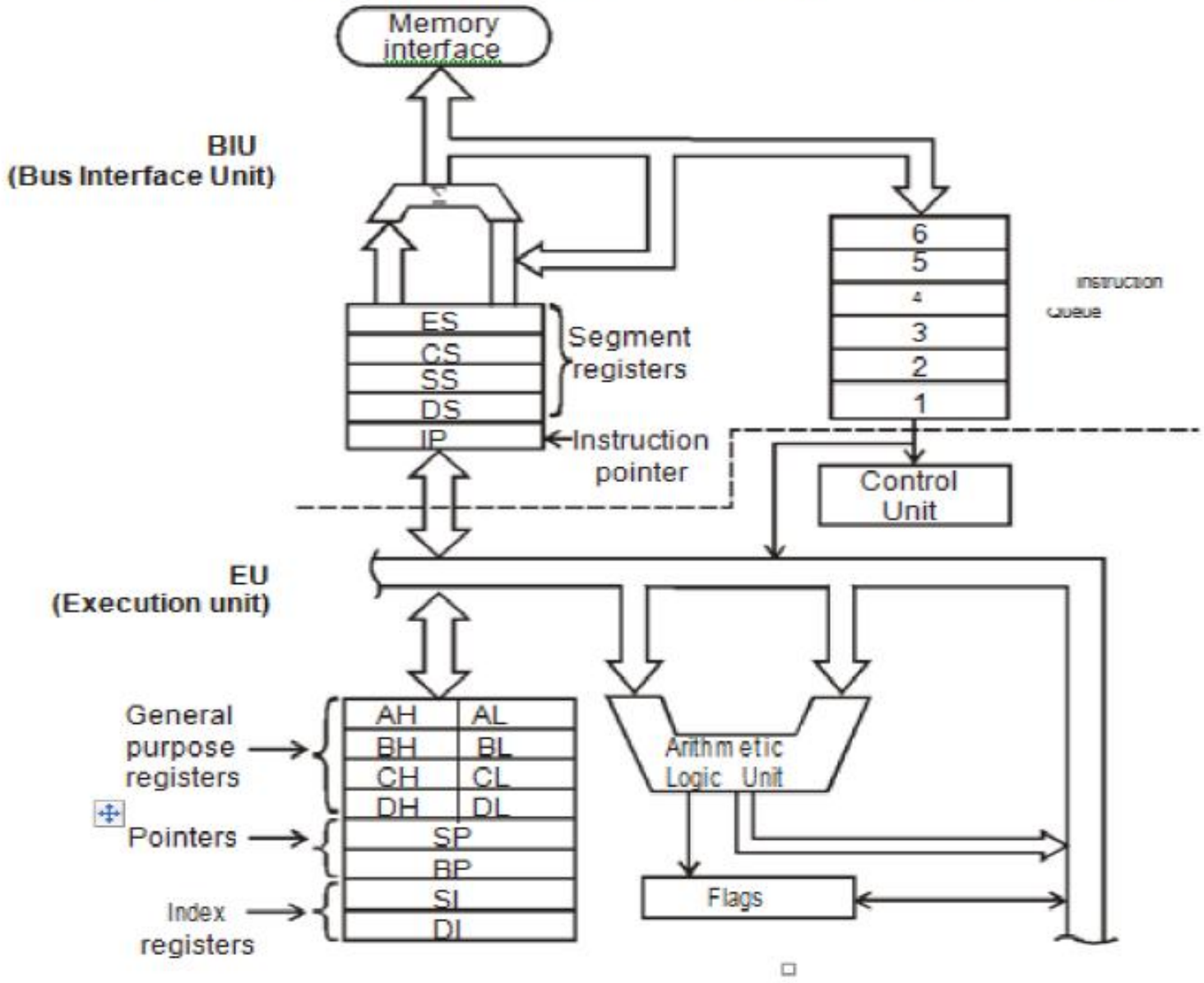
8086 Microprocessor Architecture



8086 Microprocessor Architecture

the 8086 processor are partitioned logically into two processing units

- **Bus Interface Unit (BIU)**

  The BIU fetches instructions, reads data from memory and ports, and writes data to memory and I/O ports.

- **Execution Unit (EU)**

  EU receives program instruction codes and data from the BIU, executes these instructions and stores the results either in the general registers or output them through the BIU.

8086 Microprocessor Architecture

The BIU contains

- Segment registers
- Instruction pointer
- Instruction queue

The EU contains

- ALU
- General purpose registers
- Index registers
- Pointers
- Flag register

Register

- A register is one of a small set of data holding places that are part of a computer processor.

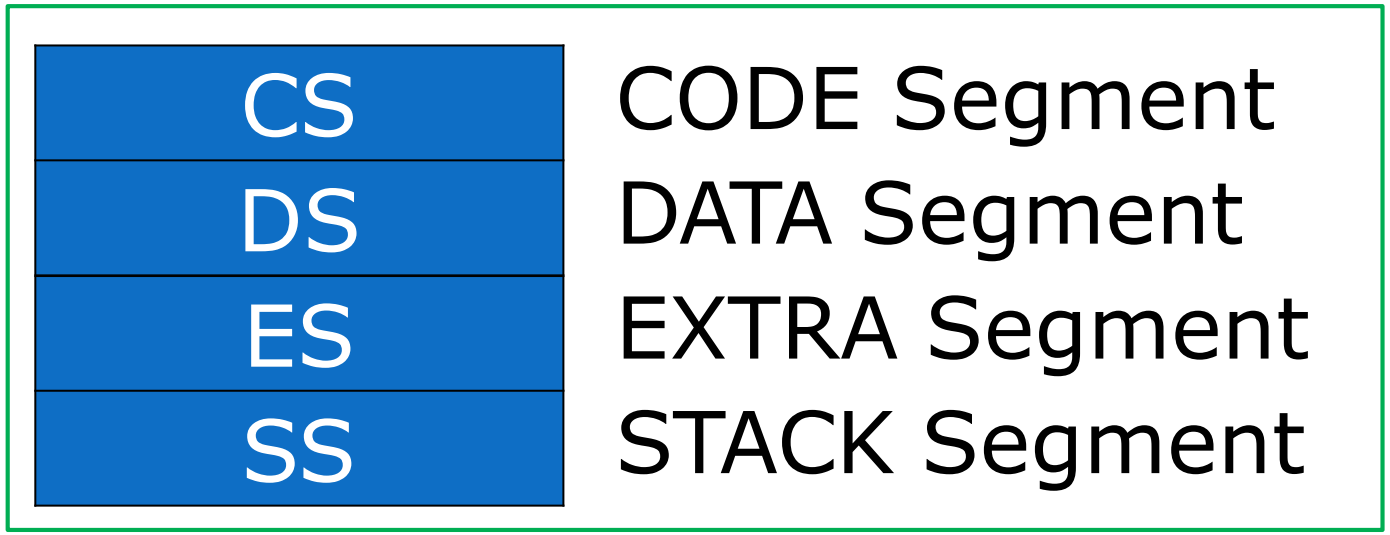
- Registers are two types:

1. General-purpose or multipurpose Registers
   - These registers hold various data sizes and used for almost any purpose.
2. Special purpose Registers
   - Used in special purposes.

Segment Register

- Segment Registers are additional registers, generate memory addresses when combined with other registers in the microprocessor.

- In a word, *segment registers are section of memory*.



Segment Register

1. Code segment (CS)

- The code segment is a section of memory that holds the code (programs and procedures) used by the microprocessor.
- The code segment (CS) register defines the starting address of the section of memory holding code.
- The code segment size is limited to max 64KB.

Segment Register

2. Data segment (DS)
   - The data segment contains most data used by a program.
   - The data segment (DS) register defines the starting address of the section of memory holding data.
   - The data segment size is limited to max 64KB.
   - Data are accessed in the data segment by an offset address or the contents of other registers that hold the offset address.

Segment Register

3. Extra segment (ES)
   - ES is an additional data segment used by some of the string instructions to hold destination data.
   - The extra segment (ES) register defines the starting address of the section of memory used as extra segment.
   - Can be used as an extra/additional segment for code or data.
   - Max size of the segment is limited to 64KB.

Segment Register

4. Stack segment (SS)
   - The stack segment defines the area of memory used for the stack.
   - The stack segment (SS) register defines the starting address of the section of memory used as stack.
   - The stack entry point is determined by the stack segment and stack pointer registers.

General-purpose or multipurpose Registers

- The MP has 07 general purpose registers.
- Each registers are 16-bit in size, while some registers can be divided into two 8-bit register.
- The acceptable 8-bit register pairs are AH-AL, BH-BL, CH-CL, DH-DL. These pairs forms AX, BX, CX, DX registers, respectively.

AX	AH	AL	Accumulator
BX	BH	BL	Base index
CX	CH	CL	Count
DX	DH	DL	Data
	BP		Base pointer
	SI		Source index
	DI		Destination index

General-purpose or multipurpose Registers

1. AX register
   - Called 16-bit accumulator, while AH/AL is 8-bit accumulator.
   - Used for instructions such as Multiplication & division.

2. BX register
   - 16-bit Base index register, while BH/BL is 8-bits.
   - Sometimes holds offset address.

Offset is the displacement of the memory location from the starting location of the segment.

3. CX register
   - 16-bit Count register, while CH/CL is 8-bits.
   - Holds count for various instructions such as SHIFT, ROTATE, LOOP, etc.

General-purpose or multipurpose Registers

4. DX register
   - 16-bit Data register, while DH/DL is 8-bits.
   - Holds a part of the result from multiplication or part of the dividend before a division.
   - Need for I/O devices.
5. BP register
   - 16-bit Base Pointer register.
   - Points to a memory location.
6. SI register
   - 16-bit Source Index register.
   - Addresses source string data for the string instructions.

Special-purpose Registers

- Special-purpose registers include IP, SP, FLAGS.

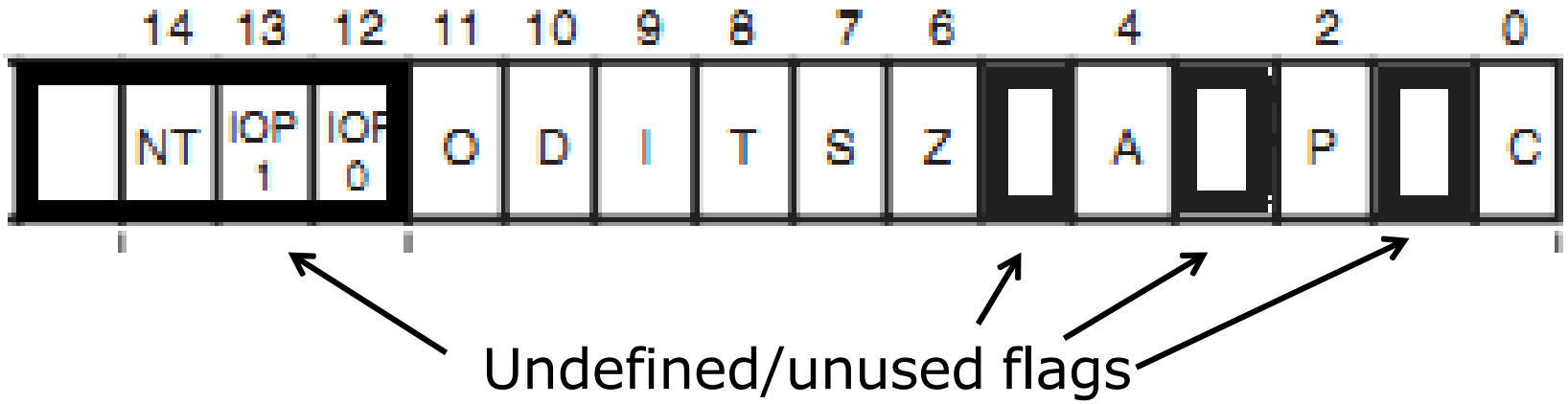
1. IP register
   - Called 16-bit Instruction Pointer.
   - The Instruction pointer, which points to the next instruction in a program, is used by the microprocessor to find the next sequential instruction in a program located within the code segment.

2. SP register
   - 16-bit Stack Pointer, addresses an area of memory called the stack.
   - The stack memory stores data through this pointer.

FLAGS register

- A flag resister (F or FL) is a 16-bit registers, which indicates some condition produced by the execution of an instruction and controls certain operation of the MP.
- Flag resister contains 09 active flags. Among them 06 are status flags (indicate some condition produced by an instruction) and 03 are control flags (control certain operation of processor). These are -
  - ✓ Carry (C)
  - ✓ Parity (P)
  - ✓ Auxiliary carry (A)
  - ✓ Zero (Z)
  - ✓ Sign (S)
  - ✓ Trap (T)
  - ✓ Interrupt (I)
  - ✓ Direction (D)
  - ✓ Overflow (O)

FLAGS register



FLAGS register

1. C (Carry) Flag
   - It holds the carry after addition or the borrow after subtraction.

FFH	**1111 1111**
+ **FFH**	**1111 1111**
FEH	**1 1111 1110**

Carry

Here, **C=1**
If No carry, **C=0**

FLAGS register

2. P (Parity) Flag

- Parity is the count of ones in register.
  - Expressed as even or odd.
  - Logic 0 for odd parity and logic 1 for even parity.
  - If a number contains no one bits, it has even parity.

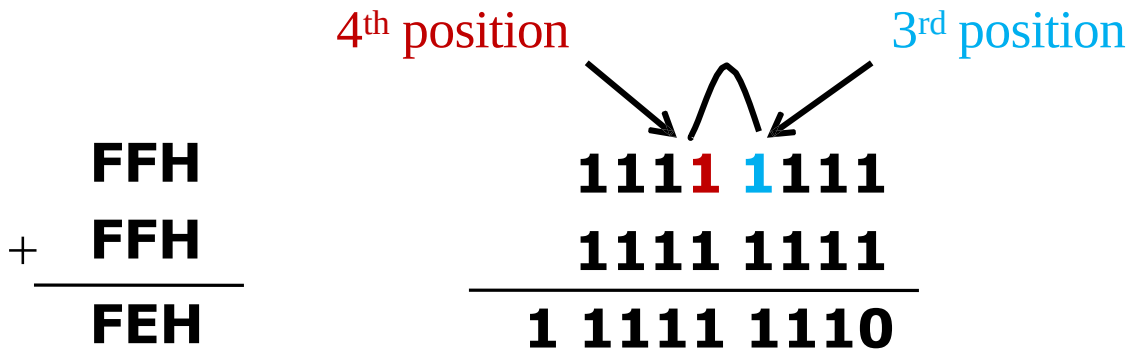
	FFH	**1111 1111**
+	**FFH**	**1111 1111**
	FEH	**1 1111 1110**

Here, No. of "1"=7,
So, Odd parity i.e., **P=0**
If even carry, P=1

FLAGS register

3. A (Auxiliary carry) Flag

- The auxiliary carry holds the carry (half-carry) after addition or the borrow after subtraction between bits positions 3 and 4.



Here, "1" transfer between 3rd and 4th positions bits. So, **A=1** otherwise, **A=0**

FLAGS register

5. S (Sign) Flag
   - The sign flag holds the arithmetic sign after an arithmetic or a logical operation.
   - If S = 1, the sign bit is set and the result is negative (-ve).
   - If S = 0, the sign bit is not set and the result is positive (+ve).
   - Usually sign bit is the leftmost bit.
6. T (Trap/Trace) Flag
   - It enables trapping through an on-chip debugging (checking error) facility.
   - T = 1, Debugging enable
   - T = 0, Debugging disable

FLAGS register

7. I (Interrupt) Flag
   - The interrupt flag controls the operations of the INTR (Interrupt request) input pin. If I = 1, the INTR pin is enabled; INTR pin is disabled with I = 0.
8. D (Direction) Flag
   - The D flag selects either the increment or decrement mode for DI and/or SI registers during string instructions.
     - If D = 1, the registers are automatically decremented; Apple
     - If D = 0, the registers are automatically incremented. elppa
   - Decrement mode indicates Set direction and write with STD (set direction) instruction. Increment mode indicates Clear direction and write with CLD (clear direction) instruction.

FLAGS register

9. O (Overflow) Flag
   - An overflow indicates that the result has exceeded the capacity of the machine.
   - Overflow occurs when signed numbers are added or subtracted.
   - Mainly occurs during multiplication operation.
     - If O = 1, Overflow occurs;
     - If O = 0, No overflow.

The flag register value after execution of an instruction is 3AD5H. Determine the status of the following flags:
Carry Flag (CY),Parity Flag (P),Auxiliary Carry Flag (AC),Zero Flag (Z),Sign Flag (S)

Memory-addressing techniques

- Memory Address is the location of a program counter in a microprocessor.
- When a microprocessor is booted up, it loads the microcode that contains the addresses of the processor’s registers.
- These registers are used to hold values that the microprocessor can read and write.
- The microprocessor places a value at the beginning of each register in the stack and places a value at the end of each register in the stack.
- When the microprocessor is called, it fetches the value at the location in the microcode that corresponds to the register on the stack.

Real Mode Memory Addressing

- Real mode operation allows MP to address only the first 1MB of memory space.
  - First 1MB of memory is called either real memory or conventional memory system.
  - DOS operating system requires MP to operate in real mode.
  - Windows does not use real mode.

Segment and Offset address

- A combination of a segment address and an offset address accesses a memory location in real mode. All real mode memory addresses must consist of segment add. plus offset address.
  - The segment address, located within one of the segment registers, defines the beginning address of any 64K-byte memory segment.
  - The offset address is also held in a register and selects any location within the 64K byte memory segment. An offset address is also sometimes referred to as displacement or logical address.
- A memory location pointed by a segment address of 1000H and offset address of 2000H is written as 1000:2000.
  - What is the actual/physical memory location for 1000:2000?

Segment and Offset address

- Calculation of physical address from segment : offset pair 1000:2000

1. Append **0H** to the right of segment value. 1 0 0 0 **0**

2. Add offset value with the appended segment value. 2 0 0 0

3. The result is the physical address for the given segment : offset pair 1 2 0 0 0

Physical address = Segment add. × 10 + Offset

If Physical address is 22004H, what will be the Segment and Offset add.?

Segment and Offset address

- Because a real mode segment of memory is 64K in length, once the beginning address is known, the ending address is found by adding FFFFH.
- If segment register contains 3000H, first address of segment is 30000H and last address is 30000+FFFF or 3FFFFH.

Default Segment and Offset Registers

- 8086 has a set of rules that apply to segments whenever memory is addressed. These rules define the segment and offset register combination.

- CS always used with IP to address next instruction in program. CS register defines start of CS and IP locates next instruction within CS.
- CS=1400H & IP=1200H, MP fetches its next instruction from memory location 15200H.

Segment	Offset	Combination	Purpose
CS	IP	CS:IP	Code / Instruction address
DS	BX,SI,DI	DS:BX/SI	Data address
ES	DI	ES:DI	String destination address
SS	SP or BP	SS:SP	Stack address

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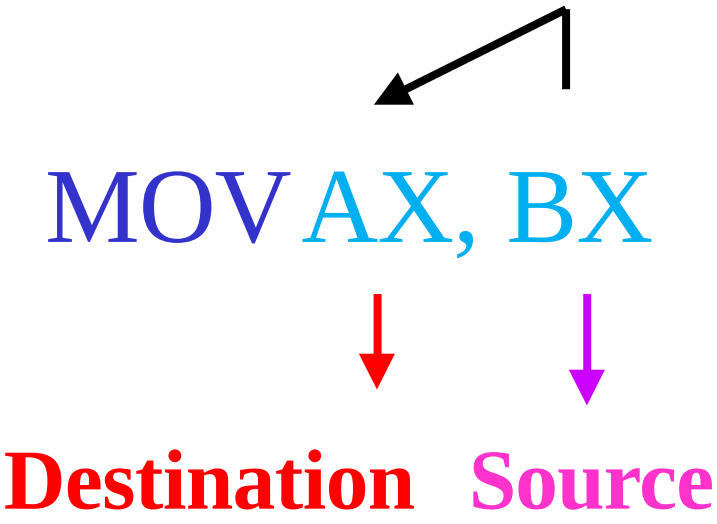
Addressing Modes of 8086

Addressing modes

- The different ways in which a processor can access data are referred t as **addressing mode**.
- Addressing modes for 8086 MP are three types –
  1. Data addressing modes
  2. Program memory addressing modes
  3. Stack memory addressing modes

Addressing modes



MOV AX, BX instruction translates the word contents of the source register (BX) into the destination register (AX).

Types of addressing mode in 8086

1. Immediate addressing mode
2. Direct addressing mode
3. Register addressing mode
4. Register Indirect addressing mode
5. Indexed addressing mode
6. Register relative addressing mode
7. Base plus index addressing mode
8. Base relative plus index addressing mode

1:Immediate addressing mode

In this type of mode, immediate data is part of instruction and appears in the form of successive byte or bytes

❖ ***Not allowed***

➢ MOV AH, 1000H

- instruction is not allowed.

MOV 75H, AL

- instruction is not allowed.

➢ MOV CS, 1000H

- Segment registers does not allow immediate data.

MOV AX,10AB$_H$

10	AB$_H$

AX

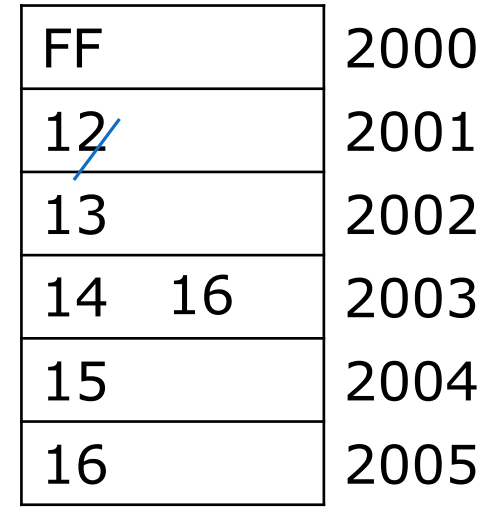
2:Direct addressing mode

3. Direct addressing

- Moves a byte or word between a memory location & a register.
- e.g., **MOV BX, LIST** instruction copies word-sized contents of memory location LIST into register CX.
- **MOV AX, [2004]** instruction copies word-sized contents of memory location 2004H into register AX.

Direct addressing

MOV BX, LIST



If,
LIST=2001H

BX=1312H

MOV DX, [2004]

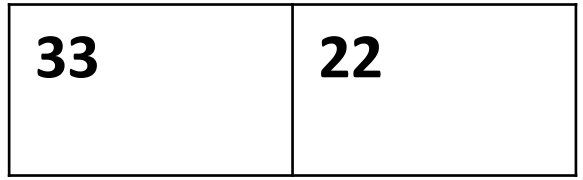
DX=1615H

MOV [2003], DH

❖ Find out BX, DX and show change in memory.

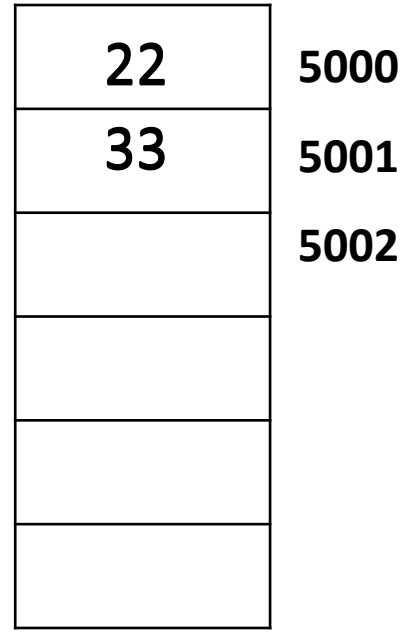
2:Direct addressing mode

MOV AX,[5000H]



AX

Memory



- ***Not allowed***
  - MOV [AX], [BX] • memory to memory transfer is not possible.

3:Register addressing mode

In this type of addressing mode, the data is stored in the register and it can be a 8-bit or 16-bit register. All the registers, except IP, may be used in this mode.

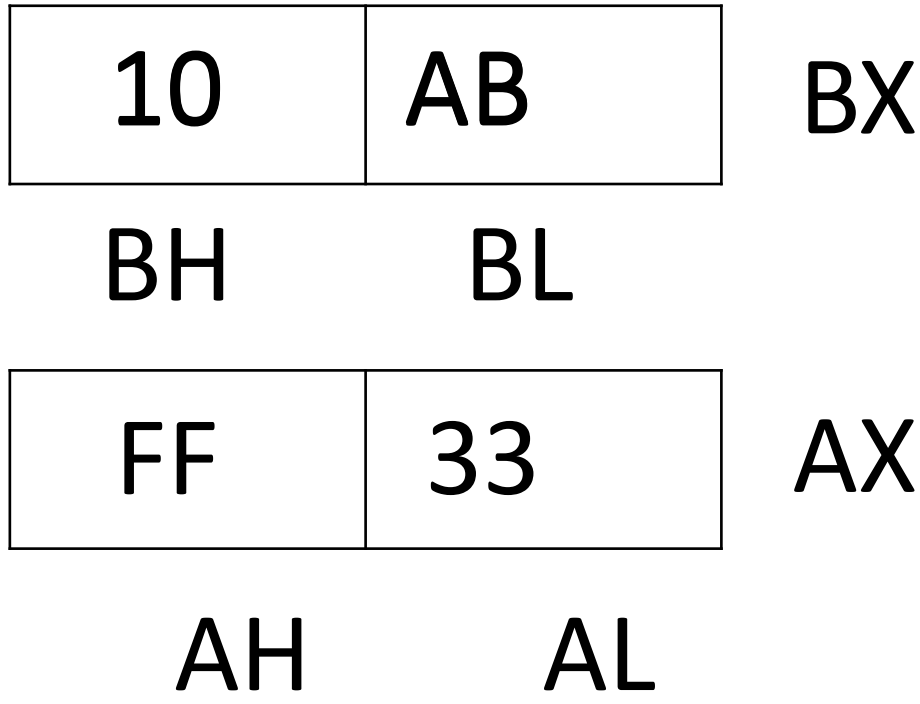
MOV AL,BLH

MOV AX,BXH

❖ ***Not allowed***

- ➢ MOV AX, AL
- ➢ MOV CS, AX
- ➢ MOV CS, DS

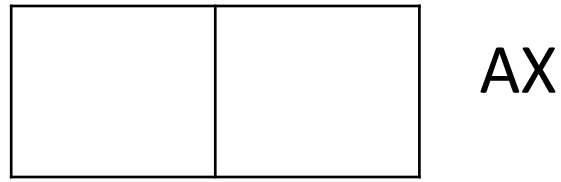
- These registers are of different sizes.
- CS can not be the destination register.
- Segment to segment transfer is not possible

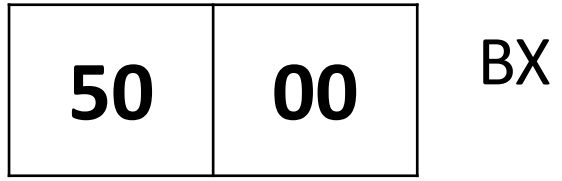


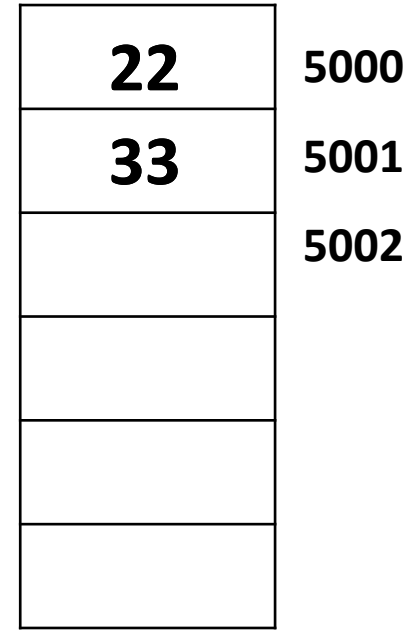
4:Register Indirect addressing mode

The address of the memory location which contains data or operand is determined in a indirect way, using the offset register.

MOV AX,[BX]







4:Register Indirect addressing mode

MOV BX, 2002H

MOV AX, 2233H

MOV [BX], AX

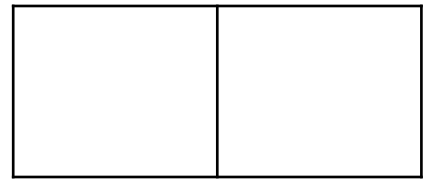
- ***Not allowed***
  - MOV AX, CS
  - MOV AX, [CS]

•CS memory itself.

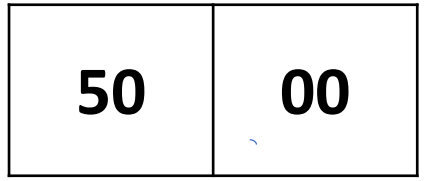
5:Indexed addressing mode

In this addressing mode, offset of the operand is stored in one of the index registers. DS is the default segment for index register SI and DI.

MOV AX,[SI]

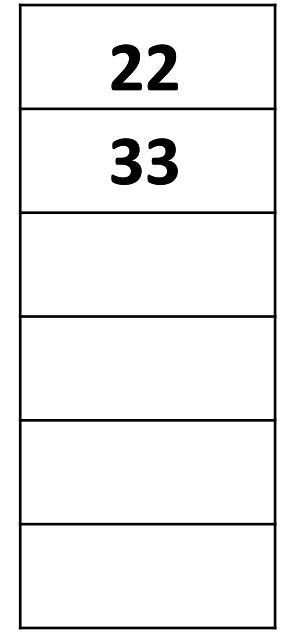


AX



SI

Memory



5000

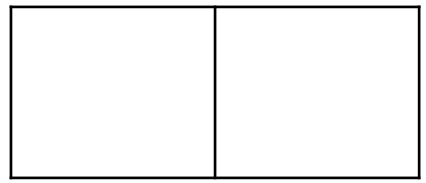
5001

5002

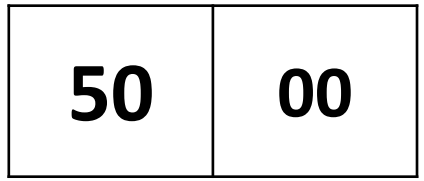
6:Register relative addressing mode

In this mode, the data is available at an effective address formed by adding an 8-bit or 16-bit **displacement** with the content of any one of the registers BX, BP, SI and DI in the default (either DS or ES) segment.

MOV AX,[BX+50H]



AX



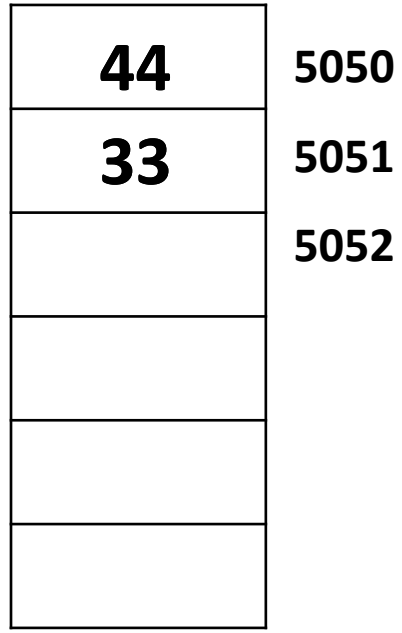
+ 50H = 5050H

BX

Offset

Final Index Address

Memory

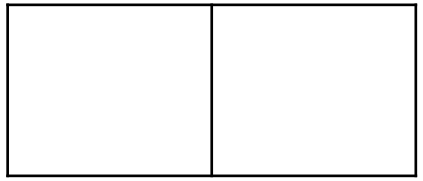


7: Base plus index addressing mode

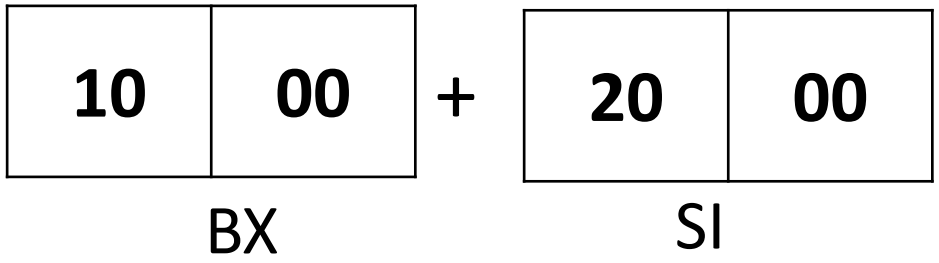
In this mode the effective address is formed by adding content of a **base register** (any one of BX or BP) to the content of an **index register** (SI or DI). Default segment register DS.

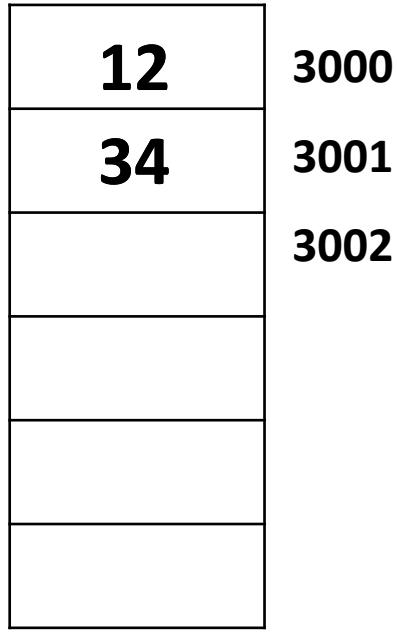
- **MOV AX, [BX+SI]** is same as **MOV AX, [BX][SI]**.

MOV AX, [BX] [SI]



AX





7: Base plus index addressing mode

MOV AX, [BX+SI]

If, BX=2003H

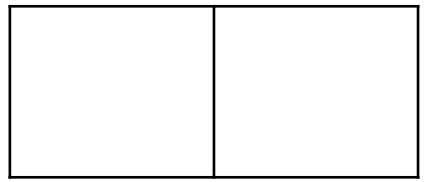
SI=0001H

AX=1615H

8:Base relative plus index addressing mode

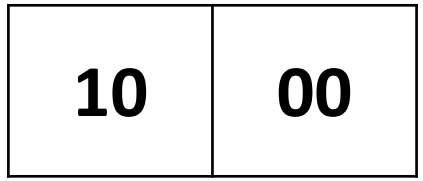
In the effective address is formed by adding an 8 or 16-bit displacement with sum of contents of any one of the **base registers** (BX or BP) and any one of the **index registers**, in a default segment.

MOV AX,50H[BX][SI]

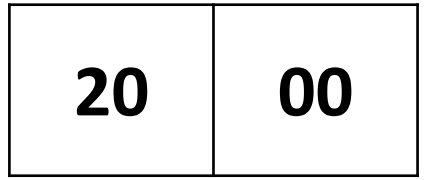


AX

50H +



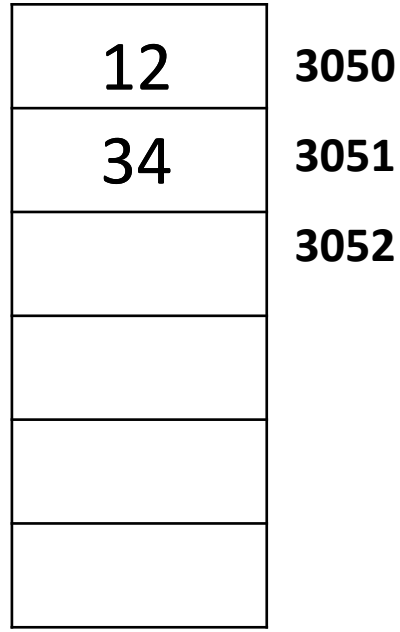
BX



SI

= 3050H

Final Index Address



Reference

Chapter 2 and 3
Barry B. Brey, **The Intel Microprocessors**